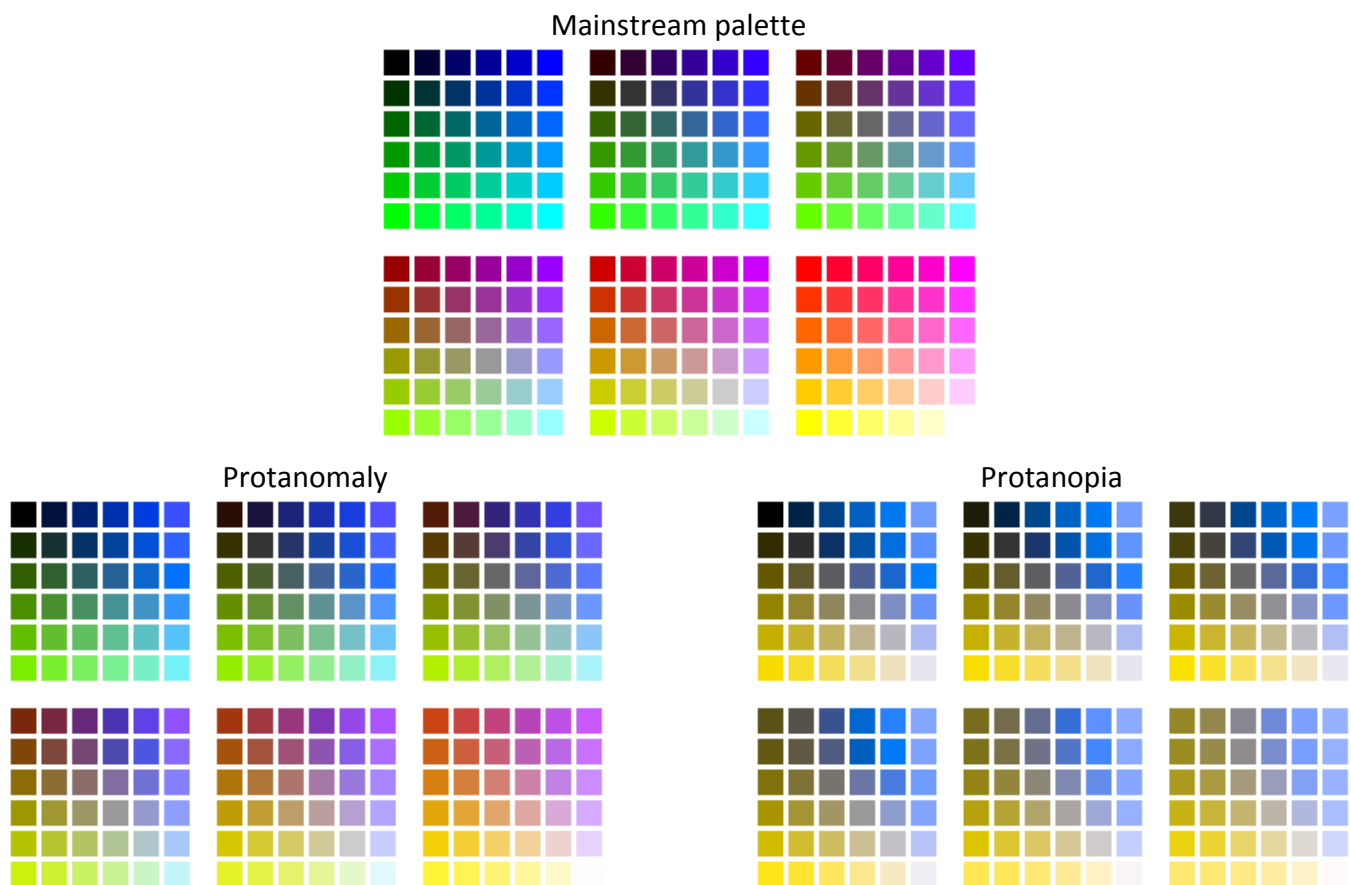


Task 3. Colour Vision

Available marks: 21

Some people have colour vision with a reduced range compared to the mainstream population. Two kinds of "colour-blindness" are called **protanomaly** (reduced red perception) and **protanopia** (severely reduced red perception). The following palettes show how the standard web-safe colours appear to someone with one of these kinds of colour vision (the differences are clearer when viewed on a monitor):



Your Task

Write a program that, given three images in PPM format:

- **R**: a *reference* palette, any palette with the layout shown above but typically the mainstream palette shown at the top;
- **T**: a *transformation* palette, any palette with the layout shown above; and
- **I**: an arbitrary image

creates a new file that shows how the third image (**I**) would look to someone whose colour vision matches the second palette **T** (or in general, the transformation implied between **R** and **T**). All image files are 8-bit RGB.

continued over...

A pixel from image **I** that has exactly the same colour as a colour in **R** must be rendered with the corresponding colour in **T**. If the colour of an image pixel does *not* appear in **R** then the nearest reference colour should be selected, using the following distance metric for the difference between any two colours A and B:

$$\text{distance}(A,B) = (\text{red}(A) - \text{red}(B))^2 + (\text{green}(A) - \text{green}(B))^2 + (\text{blue}(A) - \text{blue}(B))^2$$

where *red(C)*, *green(C)* and *blue(C)* are the red, green and blue colourants of colour C respectively.

If more than one reference colour is of equal distance then you may choose from any of these.

Palette Geometry

In the palettes shown above the coloured squares are 15px by 15px (px = pixels). The narrower white gaps between the squares are each 3px wide and the wider white gaps are 16px wide. There are no other margins, so the three palettes are exactly 347px (18*15+15*3+2*16) wide by 226px (12*15+10*3+16) high.

All palettes have 216 different colours.

Testing and Assessment

For ease of use the three palette files are pa10.ppm (mainstream), pa11.ppm (protanomaly) and pa12.ppm (protanopia).

Files containing test images are named task3a.ppm, task3b.ppm etc. Some of them will show obvious anomalies when viewed with various palettes. You will be asked to show an original image together with the result of applying a specified palette transformation by running the program with particular files.

Up to **15 marks** are available for correctly displaying images with colours that all lie within the reference palette. The remaining **6 marks** are available for correctly using the distance formula to find and use the nearest non-exact colour.